Weather-Conditioned Deep Learning for Bird Migration Trajectory Forecasting

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Abstract

Forecasting animal migration from GPS traces is difficult because most daily observations are stationary or short stopover movements, while a small fraction are long directed flights. We study next-day southbound bird trajectory forecasting from recent movement history and optional weather covariates. Our final experiment compares direct displacement regression against a triline decomposition into fly/no-fly, distance, and heading; compares Direct MLP, Triline LSTM, and Triline Transformer V2 models; and evaluates context lengths $k \in \{7, 14, 30\}$ under a pure chronological 80/20 train/test split by target date. At $k = 30$, Triline Transformer V2 reaches the best mean, median, and P90 errors in both feature settings. Without weather, Triline Transformer V2 is the strongest learned model by mean error for all context lengths. Weather improves the LSTM at shorter contexts and improves some rollout settings, but its effect remains model-dependent. Autoregressive rollout experiments show that one-step errors compound substantially over full migration paths.

1 Introduction

Bird migration prediction is a spatiotemporal forecasting problem with direct ecological and engineering relevance. Given a short window of daily GPS observations for an individual migration path, the goal is to predict the bird's next-day latitude and longitude. Accurate forecasts can support migration ecology, risk assessment for human infrastructure, and analysis of how environmental conditions shape large-scale movement. The task is challenging because migration paths alternate between stopovers and long flights: a model must decide whether the bird will move, how far it will move, and in which direction.

This project studies southbound migration trajectories with aligned weather observations. We make three empirical contributions. First, we compare direct coordinate regression with a triline movement decomposition that predicts fly/no-fly, distance, and heading before reconstructing coordinates. Second, we evaluate whether recurrent and attention-based encoders improve over a feed-forward direct baseline for short-to-medium context windows. Third, we test whether weather covariates improve generalization when train and test examples come from different years.

Our main finding is that architecture choice depends on the evaluation setting. Under chronological splitting, Triline Transformer V2 is the strongest learned model at $k = 30$. The triline LSTM remains close, converges quickly, and gives reliable rollout behavior. Weather features help in several settings but do not uniformly improve every architecture, so environmental conditioning should be interpreted as useful but not automatically beneficial.

Preprint.

2 Related Work

Animal movement forecasting builds on both ecological modeling and sequence learning. Large tracking repositories such as Movebank organize animal GPS records, individual metadata, and study structure, making multi-year trajectory analysis feasible at scale Kranstauber et al. (2011). Classical movement ecology often models paths probabilistically rather than as supervised prediction; dynamic Brownian bridge movement models, for example, estimate utilization distributions from observed tracks and time-varying movement variance Kranstauber et al. (2012). These methods are interpretable and biologically grounded, but they are not designed to learn nonlinear effects from high-dimensional temporal covariates.

Deep sequence models offer a complementary approach. Recurrent networks such as LSTMs encode ordered history with a strong recency bias, which is attractive for next-step movement prediction. Transformers replace recurrence with self-attention and have been widely adapted to long-horizon time-series forecasting. Informer introduced efficient sparse attention for long sequence forecasting Zhou et al. (2021), while Autoformer combined decomposition with attention-like autocorrelation mechanisms Wu et al. (2021). These models motivate our triline Transformer model, but our setting differs in two ways: the data are smaller and noisier than common benchmark time series, and the output distribution is highly imbalanced between stationary days and migration flights. Our triline formulation explicitly targets that imbalance by separating movement occurrence, magnitude, and direction.

3 Data

We use a cleaned southbound bird migration dataset with matched weather features. Each row is a daily path observation. The final dataset contains 13,550 daily records, 135 migration paths, and 71 source birds spanning 2013-07-30 to 2023-12-26. For each context length k, a training example consists of the previous k daily observations, and the target is the next-day GPS location.

Table 1: Dataset and split summary.

Field	Value
Daily records	13,550
Migration paths	135
Source birds	71
Date range	2013-07-30 to 2023-12-26
Context lengths	$k = 7, 14, 30$ days
Train/test protocol	Chronological 80/20 split by target date
Prediction target	Next-day latitude and longitude
Fly threshold	10 km daily displacement

The no-weather setting uses location, displacement, step length, heading, day-of-year encoding, and stopover duration. The weather setting augments these with temperature, wind speed and direction, surface pressure, log precipitation, cloud cover, and boundary layer height. All feature normalization statistics are fit only on the training windows for each k.

4 Methods

4.1 Problem Formulation

For a path with daily feature vectors $\mathbf{x}_{t-k+1:t}$ and current location $\mathbf{s}_t = (\mathrm{lat}_t, \mathrm{lon}_t)$, we predict the next location $\widehat{\mathbf{s}}_{t+1}$. Evaluation uses haversine distance between $\widehat{\mathbf{s}}_{t+1}$ and the observed $\mathbf{s}_{t+1}$. We report mean error, median error, P90 error, and migration-day error for days with at least 50 km displacement.

4.2 Direct Regression

The direct models predict normalized coordinate displacement:

$$\widehat{\Delta\mathbf{s}}_{t+1} = [\widehat{\Delta\mathrm{lat}}_{t+1}, \widehat{\Delta\mathrm{lon}}_{t+1}], \qquad \widehat{\mathbf{s}}_{t+1} = \mathbf{s}_t + \widehat{\Delta\mathbf{s}}_{t+1}.$$

This target is simple and optimized directly for coordinate movement, but it does not expose whether the model understands the fly/stationary decision.

4.3 Triline Decomposition

The triline models predict three movement components from an encoded history representation $\mathbf{h}$:

$$\widehat{p}_{fly} = \sigma(f_{fly}(\mathbf{h})), \tag{1}$$

$$\widehat{d} = \exp(f_{dist}(\mathbf{h})) - 1, \tag{2}$$

$$[\widehat{\sin\theta}, \widehat{\cos\theta}] = f_{dir}(\mathbf{h}). \tag{3}$$

The model is trained with binary cross entropy for the 10 km fly label, mean squared error on log distance, and mean squared error on heading sine/cosine. At inference, distance and heading are converted into latitude/longitude displacement using a local spherical approximation. We also evaluate gated triline predictions, where predicted distance is set to zero when $\widehat{p}_{fly}$ is below the selected threshold.

4.4 Architectures

We compare three reported model families. The Direct MLP flattens the k-day input window and predicts displacement with feed-forward layers. The Triline LSTM 4L uses a 4-layer LSTM and triline heads. The Triline Transformer V2 uses a CLS token, 4 Transformer encoder layers, $d_{model} = 256$, 8 attention heads, GELU feed-forward blocks, and triline heads. Bird identity embeddings were disabled in the final runs to emphasize generalization from path dynamics rather than memorized individuals.

5 Experimental Setup

All final experiments use seed 42, CUDA, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, batch size 128, maximum 200 epochs, and early stopping patience 30. The learning rate scheduler is ReduceLROnPlateau with factor 0.5 and patience 4. The chronological split sorts all windows by target date, assigns the first 80% to training, and assigns the last 20% to testing without shuffling. The current training code uses this test partition for learning-rate scheduling, early stopping, and checkpoint selection; there is no separate validation split.

Table 2: Model sizes in the final experiment. Parameter counts vary slightly with feature mode and context length; values shown are representative for $k = 30$.

Model	Parameters without weather	Parameters with weather
Direct MLP	47K	78K
Triline LSTM 4L	2.16M	2.16M
Triline Transformer V2	3.30M	3.30M

6 Results

6.1 Next-Day Prediction

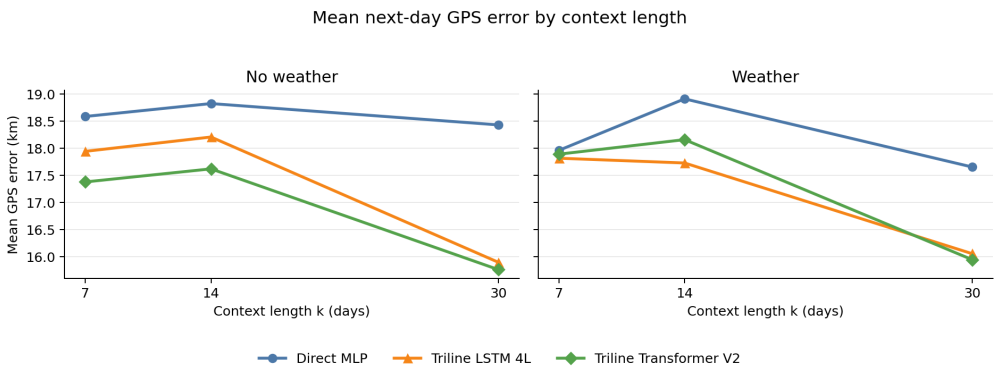

Figure 1: Mean next-day GPS error across context lengths. Triline Transformer V2 is strongest without weather across all context lengths, while Triline LSTM is strongest with weather at shorter contexts.

Table 3: Next-day prediction results for $k = 30$. Lower error is better.

Condition	Model	Mean km	Median km	P90 km	Mig.50 mean km
No weather	Direct MLP	18.43	5.26	50.98	90.07
No weather	Triline LSTM 4L	15.90	2.38	49.55	91.30
No weather	Triline Transformer V2	**15.76**	**2.31**	**49.46**	91.97
Weather	Direct MLP	17.66	4.66	51.39	95.75
Weather	Triline LSTM 4L	16.06	2.39	51.14	**89.67**
Weather	Triline Transformer V2	**15.95**	**2.28**	**45.46**	98.99

At the longest context, Triline Transformer V2 has the best mean, median, and P90 errors in both feature settings. Direct MLP remains competitive on large-movement days without weather, while Triline LSTM gives the best weather migration-day mean.

6.2 Architecture and Context Trends

Table 4: Best model by mean error for each context and feature setting.

Condition	k	Best mean-error model	Mean km
No weather	7	Triline Transformer V2	17.38
No weather	14	Triline Transformer V2	17.62
No weather	30	Triline Transformer V2	15.76
Weather	7	Triline LSTM 4L	17.82
Weather	14	Triline LSTM 4L	17.73
Weather	30	Triline Transformer V2	15.95

The chronological split favors models that can exploit recent temporal continuity while handling the stationary-day imbalance. Without weather, Triline Transformer V2 wins mean error for all three context lengths. With weather, Triline LSTM wins at $k = 7$ and $k = 14$, while Triline Transformer V2 wins at $k = 30$. The LSTM remains close and efficient, with best epochs 3, 5, and 5 in the weather runs compared with Transformer V2 best epochs 10, 21, and 21.

6.3 Weather Ablation

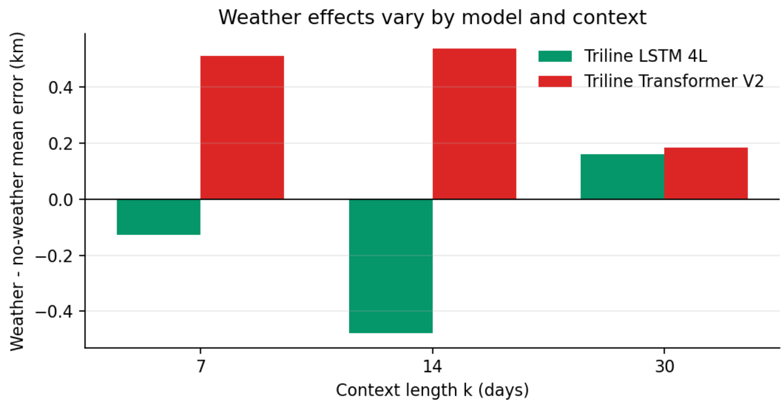

Figure 2: Weather minus no-weather mean error for triline models. Negative values mean weather helps. Weather helps the LSTM at shorter contexts but increases Transformer mean error, even while improving some tail-error slices.

Table 5: Weather ablation for triline models. Delta is weather minus no-weather mean error; negative is better with weather.

Model	k	No-weather mean	Weather mean	Delta
Triline LSTM 4L	7	17.94	17.82	-0.13
Triline LSTM 4L	14	18.21	17.73	-0.48
Triline LSTM 4L	30	15.90	16.06	+0.16
Triline Transformer V2	7	17.38	17.89	+0.51
Triline Transformer V2	14	17.62	18.16	+0.54
Triline Transformer V2	30	15.76	15.95	+0.18

Weather provides useful signal, but not uniformly. It improves LSTM mean error at $k = 7$ and $k = 14$, slightly hurts LSTM at $k = 30$, and increases Transformer mean error for all context lengths. At the same time, weather improves several tail-error slices, including Transformer P90 at $k = 30$. This suggests that weather covariates help some difficult movement cases but can also add noise or overfitting pressure depending on architecture.

6.4 Autoregressive Rollout

Next-day accuracy is necessary but not sufficient for full-path forecasting. We therefore evaluate autoregressive rollouts on 27 valid held-out paths, using at least 60 observed days before rolling forward. These rollouts repeatedly feed predictions back into the model context, so small one-step biases can grow into large path-level errors.

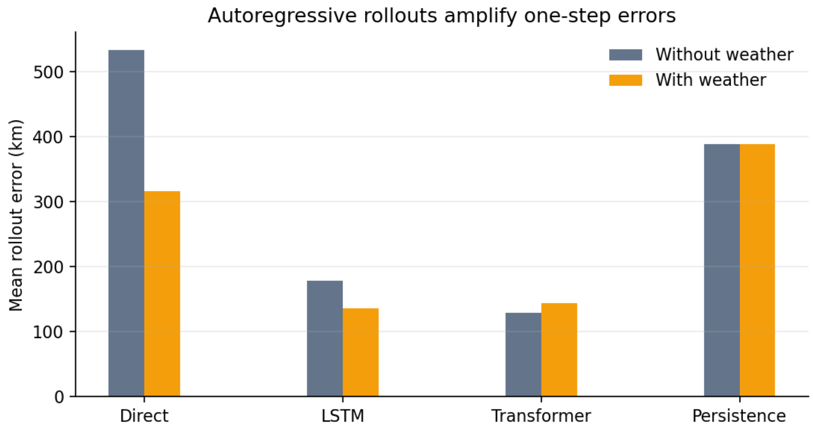

Figure 3: Mean autoregressive rollout error. Learned one-step models improve over persistence in some settings, but rollout remains much harder than one-day prediction.

Table 6: Autoregressive rollout aggregate errors over 27 valid paths.

Condition	Direct	LSTM	Transformer	Persistence
No weather mean km	534.17	178.37	**129.55**	388.62
Weather mean km	316.03	**136.61**	144.11	388.62
No weather final km	877.29	273.35	**178.28**	–
Weather final km	480.35	180.47	**166.68**	–

Rollout results are mixed. Weather substantially improves Direct and LSTM rollouts, while the no-weather Transformer has the lowest mean rollout error. Representative GIFs in `rollout/with_weather/gifs/` and `rollout/without_weather/gifs/` illustrate both successful paths and hard failures where early directional errors compound over the remainder of the migration.

7 Discussion

The direct-vs.-triline comparison shows a tradeoff. Direct MLP regression is parameter-efficient and remains competitive on some large-movement slices. However, triline decomposition gives lower mean and tail error because it models the fly/stationary imbalance explicitly. Gating further reduces typical stationary-day error, though it can hurt large migration-day accuracy when the fly head misses a departure.

The architecture comparison favors triline sequence models for the chronological one-step task. Triline Transformer V2 benefits from temporal continuity at longer contexts, while the LSTM remains competitive and converges faster, which makes it attractive when training cost or rollout stability matter.

The weather ablation is useful but nuanced. Weather improves LSTM mean error at shorter contexts and improves some rollout and tail-error metrics, but it does not uniformly improve triline Transformer mean error. Future work should use departure-time weather, tailwind and crosswind relative to intended heading, stronger regularization for weather features, and a separate validation split so test performance is not also used for checkpoint selection.

8 Conclusion

We evaluated deep models for next-day southbound bird migration forecasting using GPS history and optional weather covariates. Under a chronological 80/20 split, Triline Transformer V2 is strongest at $k = 30$, while Triline LSTM is competitive and performs well in rollouts. Weather is helpful but architecture-dependent, and full-path rollouts reveal substantial compounding error. The most promising next steps are stronger weather alignment, uncertainty-aware rollouts, and a cleaner train/validation/test protocol.

Reproducibility Checklist

Table 7: Artifacts used to reproduce the final report.

Item	Location or value
Dataset	`data/combined_southbound_paths_with_weather_matched.csv`
Final experiment script	`model_files/run_final_path_experiment.py`
Rollout/GIF script	`model_files/run_current_rollout_gifs.py`
Model definitions	`model_files/model.py`
Result summaries	`with_weather/`, `without_weather/`, `rollout/`
Random seed	42
Hardware target	CUDA GPU when available

References

B. Kranstauber, A. Cameron, R. Weinzerl, T. Fountain, S. Tilak, R. Wikelski, and R. Kays. The Movebank data model for animal tracking. *Environmental Modelling & Software*, 26(6):834–835, 2011.

B. Kranstauber, R. Kays, S. D. LaPoint, M. Wikelski, and K. Safi. A dynamic Brownian bridge movement model to estimate utilization distributions for heterogeneous animal movement. *Journal of Animal Ecology*, 81:738–746, 2012.

H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang. Informer: Beyond efficient Transformer for long sequence time-series forecasting. In *AAAI*, 2021.

H. Wu, J. Xu, J. Wang, and M. Long. Autoformer: Decomposition Transformers with auto-correlation for long-term series forecasting. In *NeurIPS*, 2021.